

Language Diversity and Typology: LIN-1050

Phonological Typology

Marina Pantcheva

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1 Terminology

- **Phonology:** The study of the sound system of languages and the universal properties exhibited by these systems.
- **Phonetics:** The study of the *physical* and *physiological* aspects of sound production and perception.
 - **Articulatory phonetics** studies how speech sounds are articulated by the vocal organs.
 - **Acoustic phonetics** studies the physical properties of speech sounds as transmitted between mouth and ear.
 - **Auditory phonetics** studies the perceptual responses to speech sounds, as mediated by the ear, auditory nerve and the brain.
- **Phoneme:** the smallest unit of sound that differentiates meaning.
- **Phone:** a concrete realization of a phoneme
- **Allophone:** a phone which is one of the set of phones which can realize a particular phoneme.

2 Vowels

Vowels are characterized according to the following *articulatory features*.

- **Height** refers to the vertical position of the tongue relative to either the roof of the mouth or the aperture of the jaw. In *high* vowels, such as [i] and [u], the tongue is positioned high in the mouth, whereas in *low* vowels, such as [a], the tongue is positioned low in the mouth.

High vowels are also called *closed*. Low vowels are also called *open* vowels.
- **Backness** refers to the horizontal tongue position during the articulation of a vowel relative to the back of the mouth. In front vowels, such as [i], the tongue is positioned

forward in the mouth, whereas in back vowels, such as [u], the tongue is positioned towards the back of the mouth.

- **Roundedness** refers to whether the lips are rounded or not. In most languages, roundedness is a reinforcing feature of mid to high back vowels, and not distinctive. Usually the higher a back vowel, the more intense the rounding.

Some languages treat roundedness and backness separately, such as French and German (with front rounded vowels).

Different kinds of labialization (roundedness) are also possible: 1) **protruded lips** and 2) **compressed lips**.

The Japanese /u/, is not rounded like English /u/, where the lips are protruded. Rather, the lips are compressed in both directions, leaving a slot between the lips for the air to escape.

Swedish and Norwegian are two of the few languages where this feature is contrastive, having both protruded-lip and compressed-lip high front vowels (/u/ and /y/).

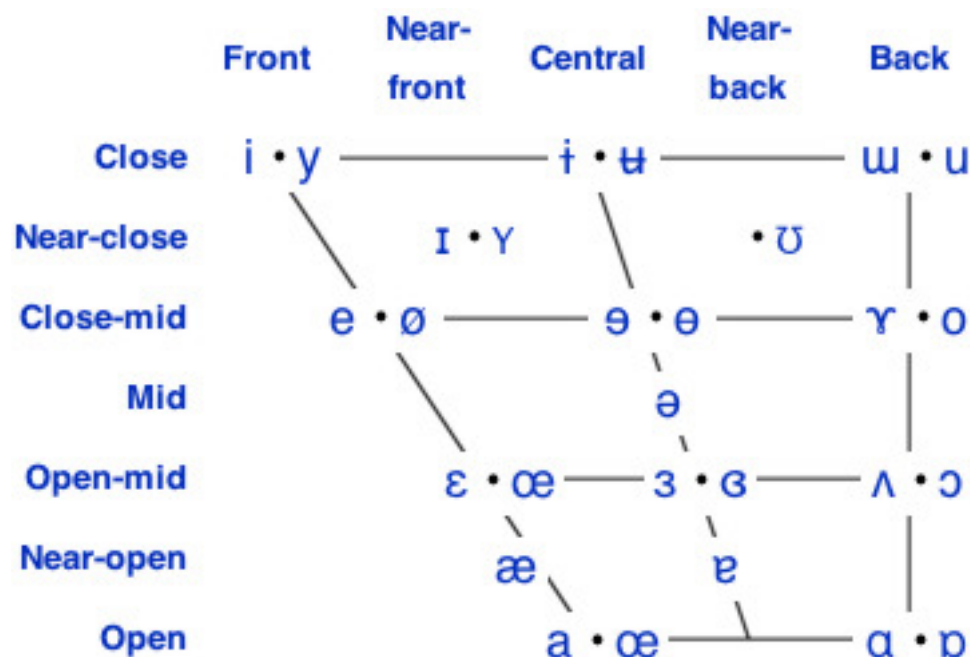


Figure 1: The schematic IPA vowel diagram

- **Nasal and oral vowels.** In nasal vowels the air escapes through the nose (and also through the mouth). In oral vowels, all air escapes through the mouth. French, Polish and Portuguese contrast nasal and oral vowels.
- **Phonation**
 - **Voicing** describes whether the vocal cords are vibrating during the articulation of a vowel. Most languages only have voiced vowels, but several Native American languages, such as Cheyenne and Totonac, contrast voiced and devoiced vowels. In

Japanese and Quebec French, vowels that are between voiceless consonants are often devoiced.

If you wonder how devoiced vowels sound, it might be helpful to know that in *whispered speech* all vowels (and also consonants) are devoiced.

- Some more “exotic” phonation types are **modal** voice, **creaky** voice, and **breathy** voice (murmured vowels). These are used contrastively in some languages.
- There are additional features as **length**, **voice onset time (VOT)**, etc.

3 Consonants

Consonant are characterized according to the following *articulatory features*.

- **Manner of articulation:** the way that the consonant is articulated, such as
 - **Plosive (oral stop):** there is complete blockage (occlusion) of both the oral and nasal cavities (/p, t, k, b, d, g/).
 - **Nasal stop** (or just **nasal**): there is complete blockage of the oral cavity, and the air passes instead through the nose (/m, n/). All languages have nasals (exception: language on Bougainville Island).
 - **Fricative (spirant):** there is continuous frication at the place of articulation (/f, s/ (voiceless), /v, z/ (voiced)). Most languages have fricatives, though many have only an /s/. However, the Australian languages are almost completely devoid of fricatives of any kind.
 - An **affricate** begins like a plosive, but this releases into a fricative rather than having a separate release of its own (English “ch” and “j”, German “pf”, “ts”). Affricates are quite common around the world, though less common than fricatives.
 - **Approximants:** speech sounds that could be regarded as intermediate between vowels and typical consonants. In the articulation of approximants, articulatory organs produce a narrowing of the vocal tract, but leave enough space for air to flow without much audible turbulence.
 - There are also **flaps**, **trills**, **liquids**, **semivowels**, etc.
 - Stops, fricatives and affricates are called **obstruents**
 - The term **sonorants** is a cover term for nasals, liquids (/r/, /l/), and approximants.
- **Place of articulation** the place where an obstruction occurs in the vocal tract between an active (moving) articulator (typically some part of the tongue) and a passive (stationary) articulator (typically some part of the roof of the mouth), and also which speech organs are involved.
 - **Bilabial:** between the lips
 - **Labiodental:** between the lower lip and the upper teeth
 - **Linguolabial consonant:** between the front of the tongue and the upper lip

- **Dental:** between the front of the tongue and the top teeth
- **Alveolar:** between the front of the tongue and the ridge behind the gums (the alveolus)
- **Postalveolar:** between the front of the tongue and the space behind the alveolar ridge
- **Retroflex:** in retroflexes, the tongue curls back so the underside touches the palate
- **Palatal:** between the middle of the tongue and the hard palate
- **Velar:** between the back of the tongue and the soft palate (the velum)
- **Uvular:** between the back of the tongue and the uvula (which hangs down in the back of the mouth)
- **Pharyngeal:** between the root of the tongue and the back of the throat (the pharynx)
- **Epiglottopharyngeal:** between the epiglottis and the back of the throat
- **Glottal:** at the glottis

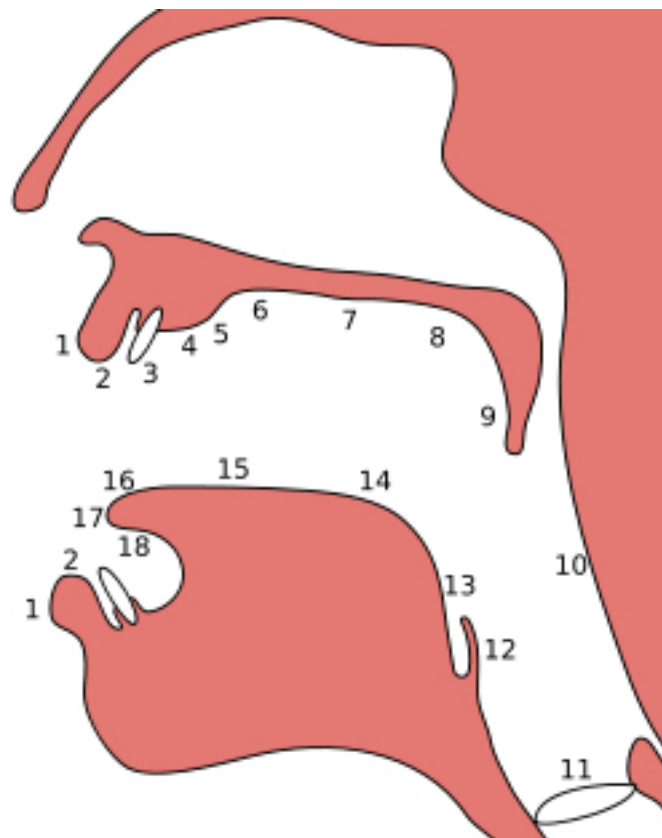


Figure 2: Places of articulation (active and passive)

- | | |
|------------------------------------|---|
| 1. Exo-labial (outer part of lip) | 4. Alveolar (front part of alveolar ridge) |
| 2. Endo-labial (inner part of lip) | 5. Post-alveolar (rear part of alveolar ridge and slightly behind it) |
| 3. Dental (teeth) | |

- | | |
|---|---|
| 6. Pre-palatal (front part of hard palate that arches upward) | 13. Radical (tongue root) |
| 7. Palatal (hard palate) | 14. Postero-dorsal (back of tongue body) |
| 8. Velar (soft palate) | 15. Antero-dorsal (front of tongue body) |
| 9. Uvular (Post-velar; uvula) | 16. Laminal (tongue blade) |
| 10. Pharyngeal (pharyngeal wall) | 17. Apical (apex or tongue tip) |
| 11. Glottal (Laryngeal; vocal folds) | 18. Sub-laminal (Sub-apical; underside of tongue) |
| 12. Epiglottal (epiglottis) | |

- **Phonation:** is how the vocal cords vibrate during the articulation. When the vocal cords vibrate fully, the consonant is called voiced (/b/); when they do not vibrate at all, it's voiceless (/b/).
- **voice onset time (VOT):** indicates the timing of the phonation. Aspiration is a feature of VOT. Many languages make a distinction between aspirated and non-aspirated consonants.
- **Airstream mechanism:** is how the air moving through the vocal tract is powered.
 - Most languages have exclusively **pulmonic egressive** consonants (in which the air stream is created by the lungs (pulmonic) exhaling and pushing air out (egressive)).
 - **Velaric ingressive:** is an airstream mechanism where a sound is produced by a closure at two places of articulation, rarifying the air in the enclosed space by lowering the tongue, and then releasing both closures. Such sounds are more commonly known as clicks.
 - **Glottalic** consonant is a consonant produced with some important contribution (a movement, a closure) of the glottis. Glottalic consonants can be:

* **Ingressive = Implosive**

How to produce an implosive consonant?

Do not try this when alone (I almost choked to death). In case something happens, call 113 (UNN hospital, Emergency)

In order to produce an implosive b, do as follows:

1. Close your lips together so as to pronounce a [b].
2. Move your glottis downward as if you were swallowing. You should be able to feel it move with your fingers; if you have a noticeable adam's apple, you should also be able to see it move in a mirror.
3. While 'swallowing', open your lips and say [ba]. Try doing this quickly so that the air flows into your mouth while you pronounce the [b]. There should be a deep hollow sound, and the [a] should follow smoothly.

The same principle applies to the other implosive consonants.

* **Egressive = Ejective**

How to produce an ejective consonant?
Be even more careful with this one!

In order to produce an ejective k, do as follows:

1. Press the back of your tongue to the roof of your mouth so as to pronounce a [k].
2. Move your glottis upward. If this is not something you normally do, you may need to monitor your adam's apple with your fingers.
3. You may notice the pressure building. Release the back of your tongue, letting out air for a [ka]. The [k] should be clicky and dull. (Your glottis will move down again during the [a], so don't mind that.)

The same principle applies to the other ejective consonants, but [k] is the easiest.

- There are some more, like **length**, but after the exercises above, nothing looks as exciting.

4 Universals of speech sounds

- The number of phonemes varies from language to language.
- Most languages have between 20 and 35 phonemes.
- Rotokas (language spoken in Papua New Guinea) has only 11.
- !Kung (spoken in Namibia and Angola) has 141!
- Some phonemes are universal, while others are found in some languages but not in others.
- There are some important absolute and statistical universals:

- (1) **All languages distinguish between vowels and consonants.**
- (2) **In a vast majority of languages the number of vowels is less than the number of consonants.**

- The only known exception is Xavante (Brazilian language), which has 13 consonants and 13 vowels.

4.1 Vowel universals

- (3) **No language has less than three vowels** (Moroccan Arabic has exactly three).
 - (4) **In languages with three vowels, there is always one front high, one back high and one low vowel.**
- Acoustically, these three vowels are maximally far removed from each other. A hypothetical three vowel system which has /i/, /y/ and /a/, for example, will be very uneconomical.

	front	back
high	/i/	/u/
low	/a/	

Table 1: Vowels in Moroccan Arabic

- (5) **No language has more than 46 vowels** (Khmer).
- The distinctive features are also universal:
- (6) **All languages make a distinction between high and low vowels.**
- (7) **All languages make a distinction between front and back vowels** (Our hypothetical /i/, /y/, /a/ vowel system would violate this universal, so good it does not exist).
- **The number of height distinctions in a system is typically equal or greater than the number of backness distinctions.**
 - **The number of height distinctions in front vowels is equal or greater than in back vowels.**
- (8) **The number of distinctions tends to be greater in the higher vowels than in the low vowels.**

	front	back
high	/i/	/u/
mid	/e/	/o/
low	/a/	

Table 2: Five-vowel system (Swahili, Spanish)

- Thus, a three-vowel system makes a front-back distinction in the high vowels (/i/ versus /u/), but not in the low vowels.
- The same can be observed in the roundedness distinction: Norwegian distinguishes three degrees of roundedness in high vowels, two in mid vowels and none in low vowels.
- The explanation for this is physiological: the more open the mouth is the more difficult it is to make a distinction along the front-back and rounded-unrounded dimensions.
- Turkish is unusual in that it makes the rounded-unrounded distinction in both high and low vowels.

	Front		Back	
	Unrounded	Rounded	Unrounded	Rounded
High	i	ü	ɪ	u
Low	e	ö	a	o

Table 3: Turkish vowel system

- Note that /e/ and /o/ fill in the slots for low vowels, while in other languages they are rather mid. This does not mean, that in Turkish /e/ and /o/ are pronounced differently.

(9) **Front vowels tend to be unrounded, back vowels tend to be rounded.**

- Exception from the universal in (9) are low back vowels, which tend to be unrounded.
- This is why, in a five-vowel system, front/mid high vowels are unrounded, back high vowels are rounded, and the low vowel /a/ is unrounded.
- **The number of vowels in a nasal vowel system is equal to or smaller than the number in the oral vowel system.**

5 Consonants

- The number of consonants varies across languages from 6 to 95 (!Xũ has 95). The average is 23.
- As we already saw in §3, consonants are characterized by combining *place* and *manner* of articulation (sometimes combined with a special feature, like *airstream initiation*, *phonation*, *nasality*).
- With respect to *place of articulation*, we have the following universals:

(10) **All languages make a distinction between labial (lips) and lingual (tongue) articulation.**

(11) **Very few languages divide the lingual consonants into coronal (front part of the tongue) and dorsal (back part of the tongue).**

(12) **While the lower lip can be combined with the upper lip (bilabial) or with the teeth (labio-dental), the distinction between bilabial and labio-dental is never utilized as a distinctive feature. It is never the only feature distinguishing two phonemes.**

(13) **Since the corona (the front part of the tongue) is the largest and the most flexible of the active articulatory organs, it provides more room for variation than dorsum (the back of the tongue) and labium (the lips)**

- With respect to *manner* of articulation, as we already saw, one basic division is between *obstruents* and *sonorants*.
- In some language, obstruents are not further divided.
- In most languages, they are further divided into two (stops and fricatives) and three (adding fricatives to them).
- Most languages divide sonorants into three: nasal, liquids and approximants.

- Further, usually obstruents are divided into two groups with respect to phonation: voiced vs voiceless, or aspirated vs unaspirated.
- The voiced-voiceless distinction is less common for sonorants.

(14) **Obstruents tend to be voiceless. Sonorants tend to be voiced.**

6 Phonotactic universals

- There are some syllable related universals:
 - (15) **All languages have syllables ending in a vowel (open syllables, no coda, (C)V), but not necessarily syllables ending in a consonant (closed syllable, coda, (C)VC).**
 - (16) **All languages have syllables with an initial consonant (onset, CV(C)), but not necessarily syllables without an initial consonant (no onset, V(C)).**
 - (17) **All languages that allow VC, allow also CVC and V, as well as the universal CV.**
- Even languages that do allow codas, sometimes have severe restrictions as to which consonants can be syllable-final. E.g. Japanese allows only a /n/ in the coda, while Thai allows /p/, /t/, /k/, /m/, /n/ and /ŋ/.
- *Consonant clusters* is another parameter along which languages vary.
- The juxtaposition of two or more consonants in a syllable is rather uncommon and many languages lack it (Turkish, Swahili, Japanese, Korean).
- Some languages allow only two-consonant clusters, but no more than that (Thai).
- Three and more consonant clusters is very rare (English *spring*, Norwegian *skjelmskt*, German *Herbsts*).
- Languages also vary in the extent to which they allow clustering of vowels into **diphthongs** and **triphthongs**.
- There seems to be no connection between vowel and consonant clusters. Chinese allows a wide variety of diphthongs and triphthongs, but no consonant clusters.